

This assignment should be done in Jupyter Notebook

1. Option Implied Volatility: Given market price of an option, solve for the volatility to match the Black-Scholes option price to the market price.

In this exercise, you will **implement a root searching algorithm using Newton-Raphson method**. Feel free to use AI to help you understand the math, but *you must write the program on your own*. You will then use this algorithm to find the implied volatility for the following options.

On Friday 10/31/2025, Palantir Technologies Inc (PLTR) closed at \$200.47.

- Its 1/16/2026 200-strike call option last trading price was \$23.50
- Its 1/16/2026 200-strike put option last trading price was \$21.00.
- The risk free rate is 3.91%
- PLTR does not pay dividends

Assuming these options are European, calculate the implied volatility of the above put and call option using Newton-Raphson.

Are the implied volatility for call and put identical? Explain why or why not.

2. In this part of the exercise, we'll assume the above options are **American** as in reality.

Implement an extended-trinomial tree, as explained in the lecture, to calculate the following Greeks of the two options in part 1. Using the previous implied volatility of the call option as input to construct the tree.

- a. Delta
- b. Gamma
- c. 1-day Theta

Show all your steps, including the system of equations you use.

3. Construct delta neutral portfolios with options and hedge shares (all options have contract multiplier = 100 shares) using the price given in part 1.
 - a. Long 1 call and short delta shares of PLTR
 - b. Long 1 put long delta shares of PLTR

Plot the PnL profile for both portfolios with varying PLTR spot price.